

Aim: Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

OUTPUT:

```

1  from cryptography.hazmat.primitives import hashes
2  from cryptography.hazmat.primitives.asymmetric import rsa, padding
3  from cryptography.hazmat.backends import default_backend
4
5
6  private_key = rsa.generate_private_key(
7      public_exponent=65537,
8      key_size=2048,
9      backend=default_backend()
10 )
11
12 public_key = private_key.public_key()
13
14 message = input("Enter the message to sign: ").encode()
15
16
17 signature = private_key.sign(
18     message,
19     padding.PSS(
20

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Tycs\cs> & C:\Users\Aqib\AppData\Local\Python\pythoncore-3.14-64\python.exe "C:\Tycs\cs\RSA Digital Signature Algorithm.py"
Enter the message to sign: AQUIB

Digital Signature Generated Successfully!
Signature (hex): 84331e7542c3508902ffc13c4d96139827d33ce0b82c1308dbdb06a1b82bc05069c44ca6d235909219196dfde6e89b53136121a446a3279db06b11895f3e1225da3ebc3dfa870e017963
6e3cd895ad14e9f5db3080e4780c3d20a336beac12c32cf50740e560799a47f06f5b2bf19a65862649526479fc3c2622c1570a3256996919b20ec882cf63ad509127c4cb88e7542f30832360dad9302086e3ac6
5c8b66fd869e08c18bd3cd407f30161e232ec78a052bdf3f5b8bd57b19350fb1dcf74c70f1f1fdb5736aed080894b2bb29c20fe5a474751ab34e779904fccaf13e30622144b3d0dff2de2ef781ce47471
6be6169927b510ad61359fc93a2

Enter the message to verify: SHA1KH

X Signature Verification failed!
The message has been modified or is not authentic.
PS C:\Tycs\cs>

```